

where n_i and n_j are the number of observations used in the estimation of Equation 13.19. The t -statistic is then calculated:

$$t_\beta = \frac{\Delta\hat{\beta}}{\hat{\sigma}_{\Delta\beta}} \quad (13.22)$$

The difference test and t -statistics for alpha follow the same form as Equations 13.20 to 13.22.

A typical t -test is conducted, which evaluates to statistically similar if the t -statistic falls into a one standard deviation interval:

$$t_\beta \in [\Delta\hat{\beta} - \hat{\sigma}_{\Delta\beta}, \Delta\hat{\beta} + \hat{\sigma}_{\Delta\beta}] \quad (13.23)$$

Complementary to the above, the difference in the alphas must show both economic viability (exceeding trading costs TC) and strong statistical significance usually at the 95% confidence interval:

$$\Delta\hat{\alpha} > TC \quad (13.24)$$

$$|t_\alpha| > [\Delta\hat{\alpha} + 2\hat{\sigma}_{\Delta\alpha}] \quad (13.25)$$

Given a pair of securities satisfying the tests of Equations 13.23 to 13.25, a long position is taken in the security with higher alpha and a short position is taken in the security with lower alpha, held and closed out by the time horizon determined in the forecast.

Index and Exchange Traded Fund Arbitrage

Index arbitrage is driven by relative mispricing dynamics of constituent securities. Recollecting that an index and exchange traded fund (ETF) is comprised of a basket of securities with the prevailing price being, usually, a weighted average of the constituent securities, the Law of One Price from economics dictates that in an efficient market, all identical goods must have only one price (Burdett and Judd, 1983). If the relative prices between constituents and their index/ETF trackers diverge an arbitrage opportunity exists. In what follows index and ETF are used interchangeably.

Exploitation of an index arbitrage opportunity where the price of the replicating portfolio net of transaction costs RP_{TC} is greater than the index price net of transaction costs I_{TC}

$$RP_{TC} \geq I_{TC} \quad (13.26)$$

would involve shorting the replicating portfolio and going long the index. This position would be closed once a profit had been realized. Similarly, where the price of the replicating portfolio net of transaction costs is less than the price of the index itself, again net of transaction costs, an